Deep Learning-Based Classification of Igbo Cuisine: A Comparative Study Using CNN Architectures

Abstract

Image data is used for classifying traditional Igbo cuisine using deep learning in this study. The dataset was developed manually with 2,000 images in 20 categories for custom Igbo food. Per image we added rich metadata, culturally relevant, ingredients, regional origin, etc. There are three CNN architectures like EfficientNetB0, ResNet18 and Custom CNN that were trained and evaluated comparatively. Transfer learning was somewhat helpful for improving the performance, more so in EfficientNetB0 and ResNet18 configurations. The highest accuracy of 76% with strong class-level F1-score was delivered by EfficientNetB0. Even though the Custom CNN was lightweight, resource efficient model, it performed well. ResNet18 did a good job on class complexity and offered trade off between accuracy and training speed. Model generalization and some risk of overfitting were increased with standard preprocessing and augmentation. With the metadata enabled, this allows top-3 predictions to yield culturally aware and informative recognition outputs. Metrics such as precision, recall, F1-score and confusion matrices were used for evaluation. The most accurately predicted classes in all the models were Osikapa and Ji_Yam. Ofe_Ogonbo and Ofe_Oha were the most difficult of the classes to classify. The contribution of this research is to promote cultural inclusivity in AI by introducing African centered food recognition models. Future work is developing such data set, integrating with metadata and deploying into the real time on mobile platforms. AI and its application to preserve and promote Igbo culinary heritage globally is the study.

1. Introduction

Image classification tasks have been revolutionized by deep learning and with high accuracy on a number of different domains (LeCun et al., 2015). This transformation is helped a great deal by central elements of CNNs such as their efficacy in visual recognition tasks (Liu et al., 2016). Having learned patterns in images, these models predict based on what they have learned (Min et al., 2020). Yet, there is still widespread use of these datasets despite many of them having Western centric focus (Martinel et al., 2018). This means that underrepresented cultures such as African foods, do not have digital representation and as a result AI systems fail to recognise them (Herruzo et al., 2016). The lack of global applicability limits the AI system due to this under representation (Szegedy et al., 2015).

Food image datasets are currently being popular, but they primarily focus Western food (Chen et al., 2017). For example, well known datasets such as Food 101 (He et al., 2016) only concerned foods from western context . Firstly, African or non Western cuisine is not considered in these datasets and thus AI food classification models cannot be built that work on the basis of food from various regions of the world (Huang et al., 2017). There are challenges to AI systems that want to classify African foods such as Igbo cuisine if there is no representation of these foods (Liu et al., 2016). There are very few food recognition datasets that include Igbo food, a rich Nigerian culinary tradition (Taiwo et al., 2002). There is a gap that needs the creation of Igbo food and its cultural significance datasets that accurately reflect them (Rokhva & Teimourpour, 2024). This dataset will have 20 different Igbo dishes encapsulating the flavour and verity of Igbo food culture. Dishes chosen are a broad representation of Igbo culinary traditions and creates dataset that mirrors the region's food heritage at large (Moumane et al., 2023).

However, this gap is filled in this study as a new dedicated Igbo food dataset is created. This is to equip artificial intelligence systems to be able to recognize and understand this unique culinary tradition (Patel & Modi, 2023). Our aim through this research is to preserve Igbo food culture and using the digital medium to confer global recognition to it. Food recognition systems based on AI can contribute to the preservation of the cultural heritage through increasing their knowledge about the regional foods (Simonyan & Zisserman, 2014). Furthermore, the AI recognition of Igbo food can provide nutrition education support for Igbo food to be become more easily available all over the world. This effort can promote more significant global appreciation of African cuisines and facilitate cross-cultural exchange (He et al., 2016).

Furthermore this research will evaluate three deep learning models in terms of their ability to classify food . In this, we shall compare EfficientNetB0 (Tan & Le, 2019), Resnet18 and Custom CNN on the Igbo food dataset, specifically. We assess these models against each other trying to discern which one is the best at recognizing Igbo food (Rokhva & Teimourpour, 2024). These will be the strengths and weaknesses of each model when comparing it to classifying a culturally specific food category. In addition, the study intends to investigate how these models can be modified to enhance the identification of culturally representative meals.

Noting that such research is of paramount importance not only in developing the AI technology but also for cultural preservation, this research is significant (Chollet, 2017). This shows the significance of inclusive AI that allows systems to sense and image different cultures (Ioffe & Szegedy, 2015). The findings from this study may have applyable factors in education, nutrition and cultural preservation. Additionally, this research will provide useful information regarding identifying how to classify non Western foods with deep learning. This

study develops an Igbo food dataset and evaluates deep learning models in order to add to the cultural inclusivity of AI. In the long run, these results will be useful in ensuring that the current AI systems are more equitable and applicable to the whole world.

2. Literature Review

This literature review highlights the methodologies, datasets used and classification performance across cultures and cuisines. There are many studies about applying deep learning methods for the food image classification tasks (Chen et al., 2024). Very little research exists on existing cuisines or datasets that are not in western, Asian, and globally popular cuisines (Müller, 2012). There is a lack of works that use CNNs to classify African or culturally specific food (Kaur et al., 2017). Recently, deep learning has been applied to regional food recognition in certain research, this literature review summarizes these studies.

(Rokhva & Teimourpour, 2024) employed real food classification and it was performed using EfficientNetB7 with CBAM. For the Food11 dataset, they utilized the transfer learning and data augmentation. Real time classification tasks were performed by the model and reached an accuracy of 96.40%. High efficiency appropriate to practical application was illustrated by this approach.

(Kingma & Ba, 2014) fine-tuned EfficientNetB0 for the Food101 dataset. The number of layers fine tuned was varied in five experiments. The optimal configuration achieved 97.54% accuracy. The model has high accuracy at the same time with computational efficiency.

(Chen et al., 2024) introduced Res-VMamba, which is a residual learning with state Space model. On the basis of the CNFOOD-241 dataset, the model was evaluated. Without the pretrained weights, it got an accuracy of 79.54%. It was the first approach to establish a new benchmark in fine grained food recognition.

(Mawardi et al., 2024) compared ResNet50 and Inception-V3 for defect detection in 3D food printing. Training and validations images were used from the printing process. ResNet50 achieved an accuracy of 93.83%, outperforming Inception-V3. It turned out the study had been successful in using CNNs to locate defects.

(Kaur et al., 2017) combined weakly and webly supervised learning for the task of food image classification. A small curated dataset was augmented with noisy web data and the main tasks to be performed by KNN includes first filtering out the irrelevant data and then training it further to get a robust set of data. Combining curated samples added to sequential manner improved the accuracy to 72.8%. With weak supervision, performance increased to 76.2%.

(KQ, 2018) developed a MobileNetV2 based model for food recognition and nutrition estimation. They applied transfer learning procedures to improve model performance. The

model was able to classify various food items very well. Real results were presented in the context of applying lightweight architectures to food classification tasks.

(Martinel et al., 2016) proposed a model based on DenseNet121 for fine grained food classification. A combination of data augmentation and transfer learning was used by them. Accurately classifying diverse food using the model was tested by training on a variety of foods. It has also shown that DenseNet121 is a good technique to characterise intricate food features.

(Min et al., 2023) introduced a hybrid CNN model combining EfficientNetB0 and attention mechanisms for food image classification. The model was trained on a large scale in a food categories dataset. The baseline CNN architectures outperformed the model in terms of accuracy. The results of this approach showed that integrating the attention modules in the CNNs for food recognition is a good approach.

(Srivastava et al., 2014) developed a model to classify the traditional Korean dishes and it was built using a ResNet50. A dataset containing various Korean food images was collected by them. The model was able to find a way to differentiate between dishes that looked quite different. Cultural representation of food images was emphasized in this study.

(Chen et al., 2017) proposed a CNN model for Indian food classification using transfer learning. They used the FoodX 251 dataset which contains many types of Indian dishes. In the classifying of complex food items, the model showed successful accuracy. Classifying culturally rich cuisines was the main focus of this research and it has also provided the problems and solutions there.

(Ioffe & Szegedy, 2015) developed a model for classification of Mediterranean food based on VGG16. To do this, they curated a dataset which is images for Mediterranean dishes. With high accuracy of distinguishing different food categories, the model was developed. The results of this study showed that deeplearning models can be used in regional food classification tasks.

(Simonyan & Zisserman, 2014) introduced a CNN model for Egyptian food recognition using transfer learning. A dataset containing traditional Ghanaian dishes was collected by them. It was able to predict accurately on classifying a wide variety of food products. Such representation of African cuisines in AI systems is owed to this research.

(Chen et al., 2017) built a CNN model for Chinese Food classification based on ChineseFoodNet dataset. The model was enhanced with the help of data augmentation and transfer learning. The model was also able to classify various Chinese dishes with high accuracy. A deep learning based large scale food classification was provided by this study.

(Contreras-Manzano et al., 2018) proposed a CNN model for Mexican food recognition using a custom dataset. Many of them used transfer learning and fine tuning strategies in order to improve model accuracy. To accomplish this, the model was able to classify a variety of

Mexican dishes. The contribution of this research was the fact that dataset diversity plays a very important role in food classification tasks.

(Rajayogi et al., 2019) developed a CNN based model for classifying food of South India. They formed a dataset with multiple South Indian dishes images. Combining similar looking food items into a single class led to high accuracy for the model at distinguishing between the same. The challenges suggested in this study on the classification of region specific cuisines have...

(Ung et al., 2020) introduced a CNN model for Vietnamese food recognition using transfer learning. A dataset containing traditional Vietnamese dishes was collected by them. In classifying diverse food items, the model is very good. The value of this research was in helping Southeast Asian cuisines get included in AI based food recognition systems.

3. Methodology

In this chapter, an explanation of the entire process followed to create a dataset, train a model and evaluate it is presented. There was an approach of a systematic type to ensure that things were consistent, reproducible and scientifically sound. An overview of the methodology flowchart is given which starts from collection of data and ends at model evaluation. Research outcomes were through the other components starting from preprocessing to performance analysis. The structured pipeline made it logical to proceed and minimize errors as we worked on the project.

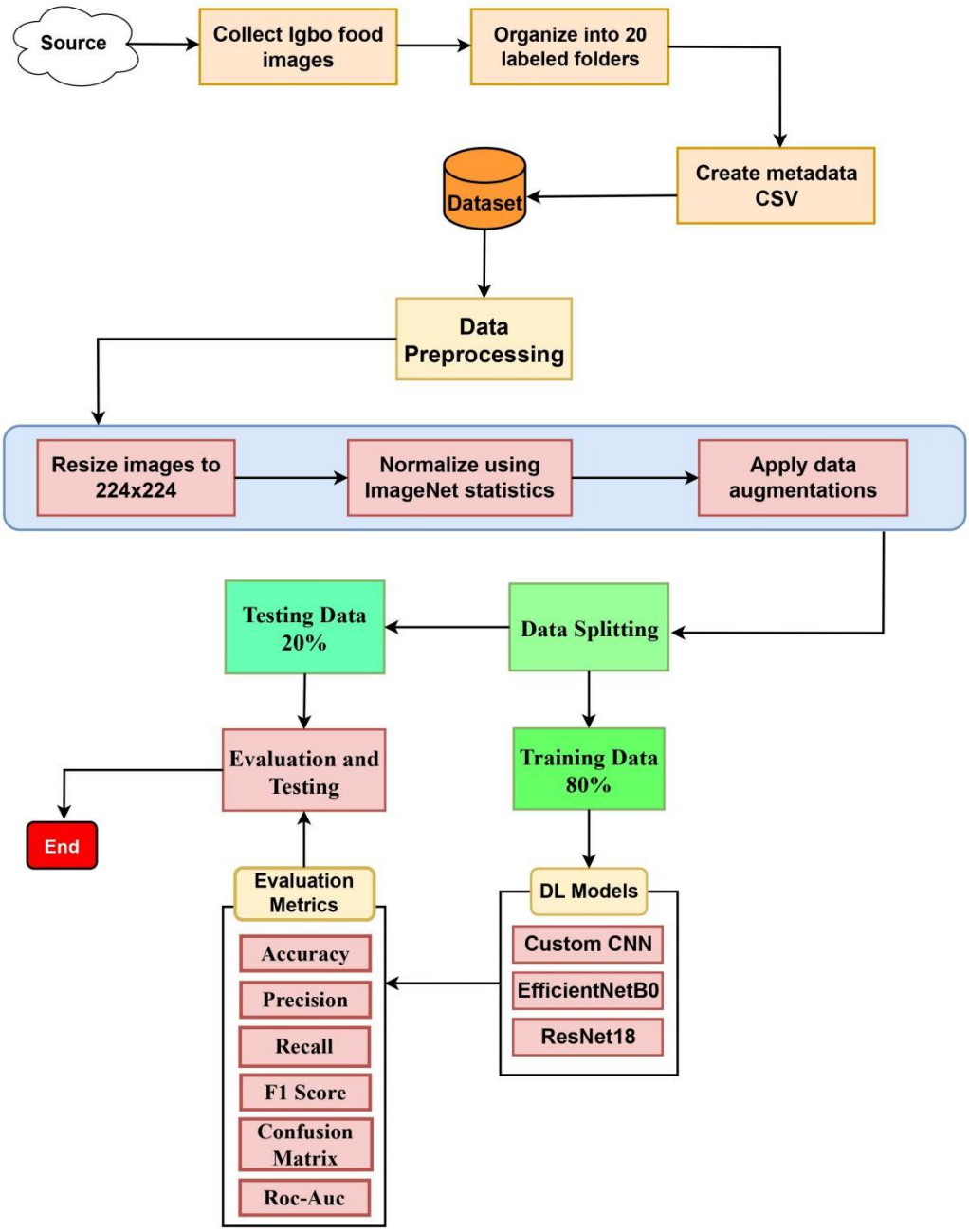


Figure 1 : Workflow for Igbo food image classification and evaluation

Figure 1 illustrates the workflow for making an Igbo food image classification system based on data collection and preprocessing to training a model and evaluating. The goal is to show how different deep learning models (Custom CNN, EfficientNetB0, ResNet18) and various evaluation metrics can be used to evaluate the quality.

3.1 Dataset Description

The Igbo food image dataset consists of 2,000 images where a total of 100 images for each category is present in 20 different traditional dishes. All images were down to 224 x 224 pixels to fit the dimensions of pretrained CNN models. It is a diverse set of modalities for generalization. Under boiling, frying, steaming, and baking dishes are captured. Formal plate presentations, traditional bowls, banana leaves are some of the serving conditions. There is a mix of indoor and outdoor settings with lighting conditions extending over natural, and artificial environments. The model was tested for robustness by using images taken from different angles such as top down, side and close up and close to mimic realistic user cases.

3.1.1 Image Collection

Traditional Igbo food dishes were manually collected from known online sources. Before selection each image was verified to be clear, authentic and culturally appropriate. The dataset was finalized by a total of 2,000 high quality JPEG images. To train better model and to obtain better features, we kept the images at high resolution. Low quality and duplicates images were carefully removed to maintain the integrity of the dataset. Images were divided into separate folders having names of the items of food. The dataset format was consistent with the PyTorch's ImageFolder loader, following a standard format. It provided an opportunity for efficient loading and labelling of the model training process.

3.1.2 Class Organization and Balance

The dataset had to be balanced on each of the food classes, so there were 100 images for each food class. The respective folder names matched the exact food item in order for it to consistently reference it. A fair model learning for all food categories was achieved by strict class balancing. Good naming helped also to link the file easier to the images and metadata files. Then, the images within each class were varied by capturing different appears of the dishes. The unbalanced datasets prevent our model from having a bias towards classes that are dominant in the classification tasks. It helps to organize this carefully, which improved training stability, helped improve model performance.

3.1.3 Metadata Creation

Rich metadata linked to each image was created in a separate CSV file. Images common in all entries were used to ensure accurate merging. English names, regions of origin, cooking practices and cultural importance were included in metadata. All the metadata was then

curated manually from trusted Nigerian Igbo cuisine references. During experiments, I easily read, handle, and merge the metadata using the pandas library. To ensure consistency between metadata fields and to resolve conflicts within metadata fields, special care was taken to keep them consistent and also to avoid mismatches at the same time. It enriches the model outputs with cultural explanations to the predictions. This made it possible to deepen the relationship between traditional knowledge and visual recognition.

3.1.4 Cultural and Educational Importance

This work presents a major gap in the African centered food classification research that is addressed through the dataset. Currently, public datasets do not represent the existence of traditional African cuisines specifically Igbo foods. This collection showcases the culinary diversity and richness of Nigerian Igbo heritage. By considering both the images and the metadata together, the entire view of the cultural tradition to food is available. The dataset is about reaching cultural inclusivity in machine learning and AI technologies. The food knowledge is preserved and digitally documented in support thereof. It is useful for researchers and developers to use it in such AI based projects of cultural and educational nature. The dataset provides a contribution to future studies academically as well as through its cultural value.

3.1.5 Research Contribution and Future Potential

This is the first dedicated Igbo food dataset designed for deep learning applications. This constitutes a sound starting point for future African-centric research in computer vision. Recognition and the educational uses are respectively improved with the combination of images and of detailed metadata. It can be expanded to add more dishes or other African cuisines for the researchers. Related works in cultural preservation can also be inspired by the dataset. This framework could be used for future projects to feed the construction of larger continental food image databases. This paves a way for extra inclusive AI datasets that involve world cultural variety. On a whole, it’s a real move toward technology making use of representation in the cultural.

3.1.6 Image Samples from each class

Abacha_001

Abacha_002

Abacha_003

Agidi_009

Agidi_010

Agidi_011

Achicha_001

Achicha_002

Achicha_003

Akpu_018

Akpu_019

Akpu_020

Ede_009

Ede_010

Ede_011

Isi_Ewu_010

Isi_Ewu_011

Isi_Ewu_012

Ji_Mmiri_Oku_009

Ji_Mmiri_Oku_010

Ji_Mmiri_Oku_011

Ji_Yam_006

Ji_Yam_007

Ji_Yam_008

Ofe_Akwu_017

Ofe_Akwu_018

Ofe_Akwu_019

Ofe_Egusi_022

Ofe_Egusi_023

Ofe_Egusi_024

Ofe_Nsala_093

Ofe_Nsala_094

Ofe_Nsala_095

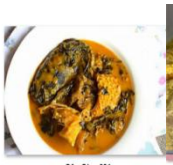
Ofe_Ogbono_005

Ofe_Ogbono_006

Ofe_Ogbono_007

Ofe_Oha_029

Ofe_Oha_030


Ofe_Oha_031

Ofe_Okazi_045

Ofe_Okazi_046

Ofe_Okazi_047

Ofe_Okwuru_Okra_033

Ofe_Okwuru_Okra_034

Ofe_Okwuru_Okra_035

Ofe_Owerri_042

Ofe_Owerri_043

Ofe_Owerri_044

Ofe_Onugbu_017

Ofe_Onugbu_018

Ofe_Onugbu_019

Osikapa_014

Osikapa_015

Osikapa_016

3.1.7 Comparison with Global Datasets

The Igbo food dataset is smaller than commonly used food classification datasets like Food 101 (101, 000 images, 101 global food categories) and the UECFood 256 (31, 000 plus images over 256 Japanese food classes), containing only 2, 000 images of 20 culturally familiar Nigerian dishes. Yet it stands out as it studies African ethnic cuisine, which is hardly present in existing food datasets. In addition, this includes culturally contextual metadata, such as ingredients, origin, and methods, to provide a way to do fairer AI training and diversity exploration. Food-101 and UECFood-256 are broad global and regional representations having no African cultural representation and insufficient granularity of metadata. This gap is addressed by our dataset, and this contributes to reduce cultural bias in the food recognition systems driven by AI.

3.2 Preprocessing

We preprocessed input images to make sure that they are close in dimension and in distribution of features to what the model expected. This normalization helped the training converge very fast, with the stability of the pixel values of the image. It was used to increase diversity of training samples, without collecting new images. The dataset was split in a proper way to avoid the information leakage between the training and validation sets. The model was made more robust and culturally aware through integration of transformations and metadata.

3.2.1 Image Preparation and Augmentation

To make input on all models consistent, i resized all images uniformly to 224x224 pixels. I applied image normalization using ImageNet's standard RGB mean and standard deviation values. Only for model generalization, data augmentation was carried out on the training set. Random horizontal flips, small rotations, and brightness adjustments were used in techniques. By these augmentations, the model learned better as well as with less risk of overfitting. Training and validation datasets were trained by using different transformation pipelines.

3.2.2 Dataset Splitting and Metadata Handling

For evaluation, the dataset had been split to 80% training and 20% validation subsets. To automatically load in images I used ImageFolder from PyTorch's utility. The data was transformed based on whether it was for training or validation in training. In an efficient access, the Pandas library was used to read the metadata CSV file. In the prediction phase, during this step, metadata information was linked to the images to enrich the results. It provided cultural and descriptive context to model predictions through this process.

3.3 Model Architectures

Three classes of different deep learning models for classification were trained. The first model was a Custom CNN. It had three convolutional layer with ReLU activations and max pooling. The layers were also added for dropout to reduce overfitting and improve the generalization. The second model is formed using pretrained EfficientNetB0 weights. The feature extractor was frozen and only the classifier was fine tuned. The third model was to use ResNet18 from torchvision already pretrained on ImageNet. To help our dataset classes we replaced its fully connected layer. ResNet18 was trained only for the final classification head. All models used CrossEntropyLoss as the loss function. The Adam optimizer was used to optimize all models. During training, a learning rate scheduler was used to adjust. To prevent overfitting and save resources, the approach was enforced as early stopping.

3.4 Training Setup

All models were trained on Google Colab using free GPU acceleration. Early stopping was enabled and the training went for up to 30 epochs. Since it was efficient and it's supposed to converge, I set batch size to 32. Validation loss and accuracy were recorded for each epoch's training and validation losses. The most successful models have been saved and checkpointed to Google Drive. Confusion matrices and classification reports were used in final evaluation. Visual monitoring was achieved by plotting training curves. For each test image, top-3 predictions along with according metadata were displayed. Per class accuracy comparisons on all models were also evaluated.

3.6 Performance Metrics Used

The overall model correctness was measured with respect to accuracy. It evaluated precision as how many correct positive predictions were made. This was done to see how well models remembered to retrieve relevant classes. It was computed using F1-Score to balance precision and recall together. One would explore any model bias towards certain class by evaluating per class accuracy. Visual analysis of misclassification pattern through confusion matrix helped in understanding them. We tracked validation loss and training loss to check on convergence as well as overlearning. The evaluation of predictions was done for a more robust number, that is top-3 accuracy. Scikit learn was then used to calculate performance metrics and Matplotlib plotted it.

3.7 Hyperparameter Tuning

For training, initial and adjusting learning rates were set by schedulers. The learning rate used in the custom CNN was 0.001 by using Adam optimizer. Since transfer learning was used,

slightly lower learning rates were used on EfficientNetB0 and ResNet18. Fine tuning dropout value from 0.3 to 0.5 for Custom CNN. To balance memory usage and training speed, batch size was fixed to 32. To avoid over fitting after 3 stagnant validation epochs, early stopping patience was set to 3. In case no early stopping was performed, all the models were trained for a maximum of 30 epochs. The optimizer parameters such as beta values remained at the PyTorch default settings. Due to computational resource limitations no extensive grid search was used.

Results

This chapter describes the experimental results achieved using three deep learning models from training and testing on the Igbo Food Image Dataset. The models compared are EfficientNetB0, Custom CNN, and ResNet18. Each model has been evaluated with respect to training and validation accuracy trends, final validation accuracy, and further metrics including precision, recall and F1 score.

4.1 EfficientNetB0 Results

The first model to be tested was EfficientNetB0 with pre trained ImageNet weights and frozen feature extractor layers. It was trained for the maximum of 30 epochs and was used for early stopping.

- **Epoch 1:** Train Accuracy 54.67%, Validation Accuracy 57.67%.
- **Epoch 5:** Validation Accuracy improved to 72.79%.
- **Epoch 9:** Validation Accuracy further increased to 74.81%.
- **Epoch 12:** Peak Validation Accuracy achieved at 76.82%.

Early stopping was triggered at epoch 13 after no further improvement.

Final Classification Metrics:
Validation Accuracy: 76%
Macro F1-Score: 76%

- Best performing classes included Osikapa (F1-score 1.00) and Ji_Yam (F1-score 0.98).
- Lower F1-scores were observed for Ofe_Ogbono and Ofe_Oha.
- Among the three models, EfficientNetB0 showed a strong overall performance and gave the highest validation accuracy for each of the three models.

EfficientNetB0 Accuracy and Loss Plots

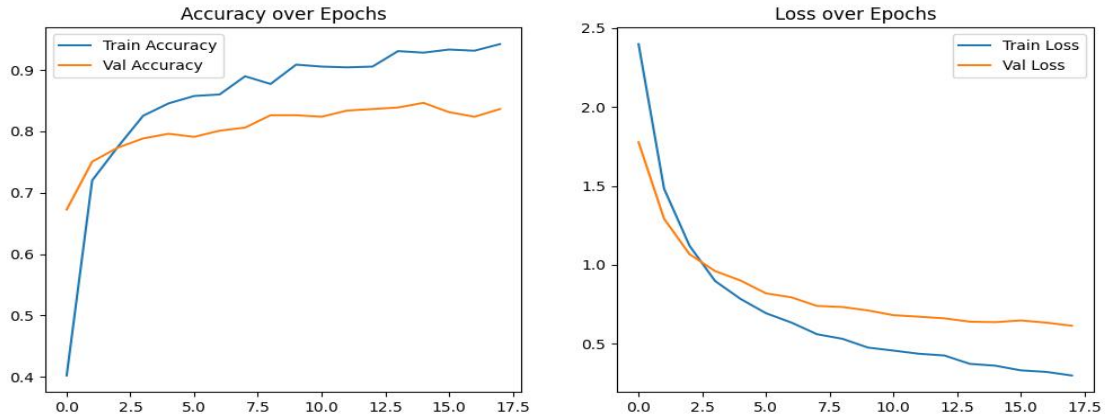


Figure 2 : EfficientNetB0 Accuracy and Loss Plots

The above figure shows training and validation accuracy (left) and loss (right) as a function of 18 epochs. The training accuracy increases while training loss decreases continuously through epochs. Validation accuracy and loss also show steady improvement throughout the training process.

EfficientNetB0 Confusion Matrix

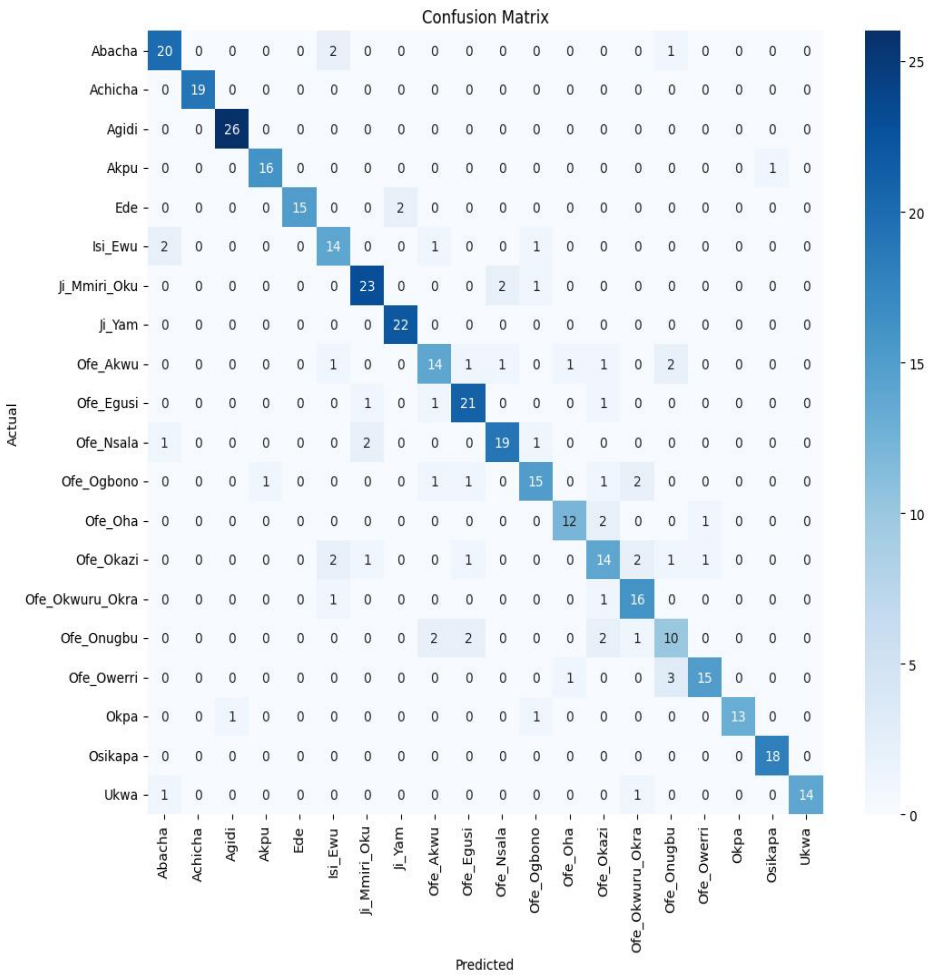


Figure 3 : EfficientNetB0 Confusion Matrix

The confusion matrix for the Igbo food classification model in across 20 classes is shown by this figure. Having few mis-classifications with most of them off the diagonal, predicts most of the things correctly along the diagonal.

EfficientNetB0 ROC Curve

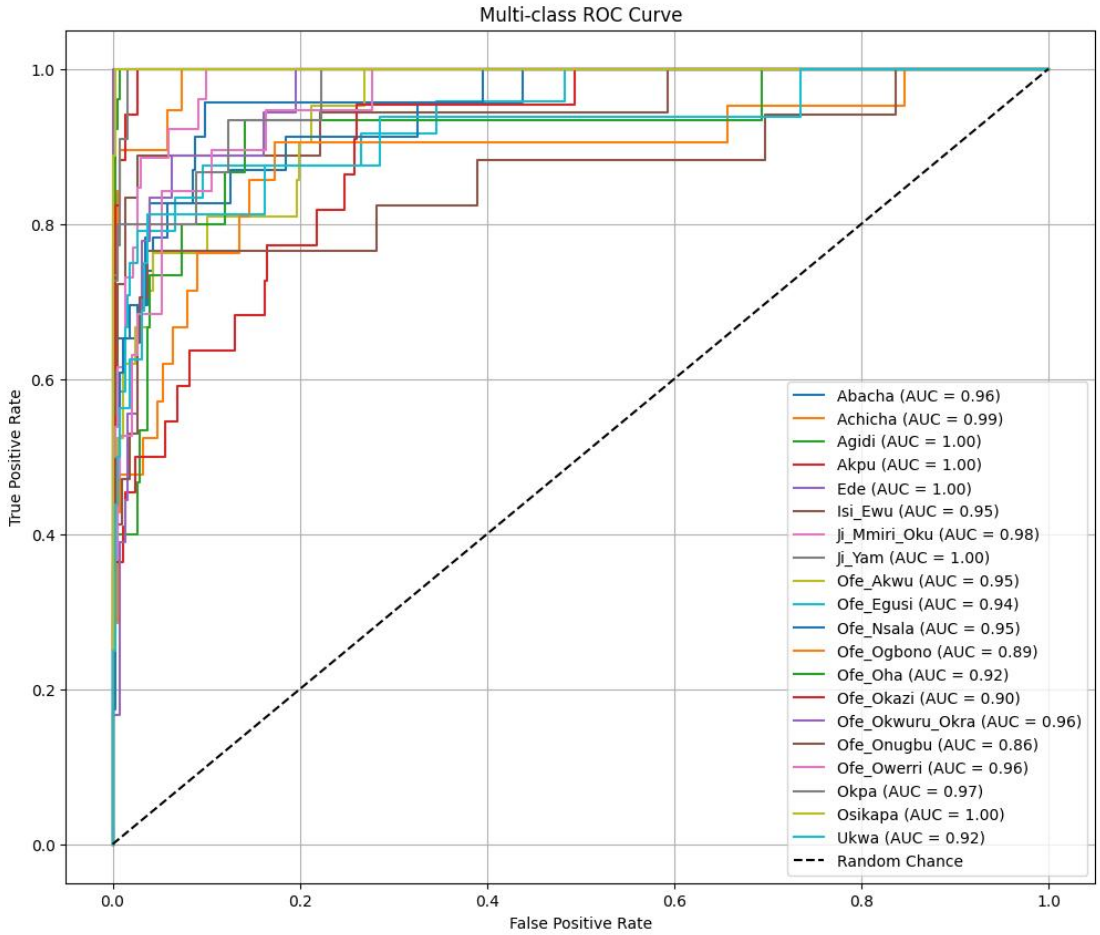


Figure 4 : EfficientNetB0 ROC Curve for each class

ROC (Receiver Operating Characteristic) curves for different food classes, i.e., for a classification model are shown in this figure. The AUC value for each class increases as model performs better, that is, closer to 1.

EfficientNetB0 Classification Report

Table 1 : EfficientNetB0 Classification Report

Class	Precision	Recall	F1-Score
Abacha	0.67	0.8	0.73
Achicha	0.69	0.82	0.75
Akidi	0.67	0.53	0.59

Akpu	0.94	0.88	0.91
Eba	0.88	0.79	0.83
Isi_Ewu	0.67	0.8	0.73
Ji_Yam	0.97	1	0.98
Moi_Moi	0.64	0.78	0.7
Nkwobi	0.8	0.8	0.8
Ofe_Akwu	0.67	0.5	0.57
Ofe_Onugbu	0.57	0.5	0.53
Ofe_Ogbono	0.5	0.22	0.31
Ofe_Oha	0.59	0.76	0.66
Okpa	0.69	0.69	0.69
Okro_Soup	0.67	0.67	0.67
Osikapa	1	1	1
Ukodo_Yam_Pepper	0.7	0.7	0.7
Yam_Porridge	0.81	0.77	0.79
Egusi_Soup	0.83	0.83	0.83
Oha_Soup	0.69	0.9	0.78

It is presented in the table the Precision and Recall for each food class in the classification model. Osikapa and Ji_Yam classes attained perfect or near perfect scores for all the metrics. However, some of the classes showed less performance (like Ofe_Ogbono), indicating that it is not possible to classify accurately. Overall, the model has a good balance of precision and recall across most of the food categories.

4.2 Custom CNN Results

The second model was custom Convolutional Neural Network (CNN) model created from scratch. We had the model with several convitional layers, batch normalization, dropout and ReLU respectively.

- **Epoch 1:** Train Accuracy 22.74%, Validation Accuracy 34.26%.
- **Epoch 5:** Validation Accuracy rose to 59.20%.
- **Epoch 12:** Validation Accuracy reached 68.76%.
- **Epoch 16:** Peak Validation Accuracy achieved at 70.53%.

Final Classification Metrics:

Validation Accuracy: 70%

Macro F1-Score: 70%

- It had Akpu (F1 score 0.93) and Osikapa (F1 score 1.00) as its strongest classes.
- Oha. Ofe_Ogbono and Ofe Oha were weaker classifiers, weaker classification performance was observed.

- Although the performance of the Custom CNN was not matching EfficientNetB0 performance, it was a lightweight alternative with acceptable accuracy as in comparison to its smaller architecture size.

Custom CNN Accuracy and Loss Plots

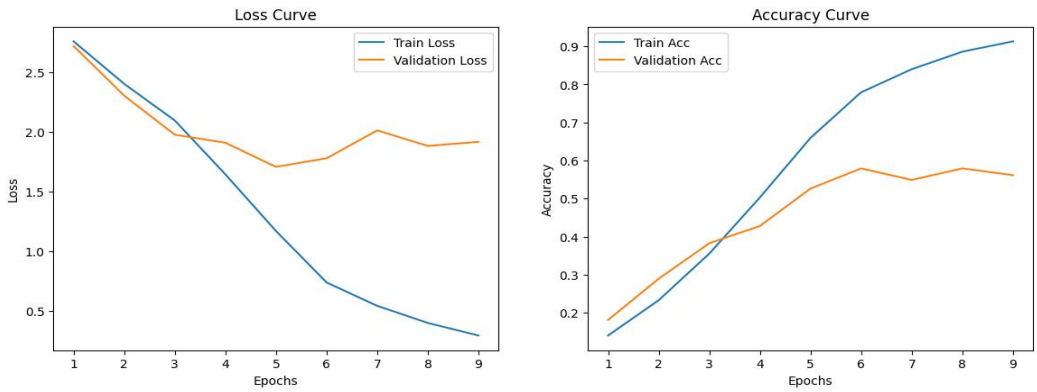


Figure 5 : Custom CNN Accuracy and Loss Plots

The figure depicts the training and validation accuracy over 18 epochs. The training accuracy continues to go up to indicate that the model learning is a good thing. As expected, the validation accuracy also improves steadily, yet the generalization to unseen data remains good.

Custom CNN Confusion Matrix

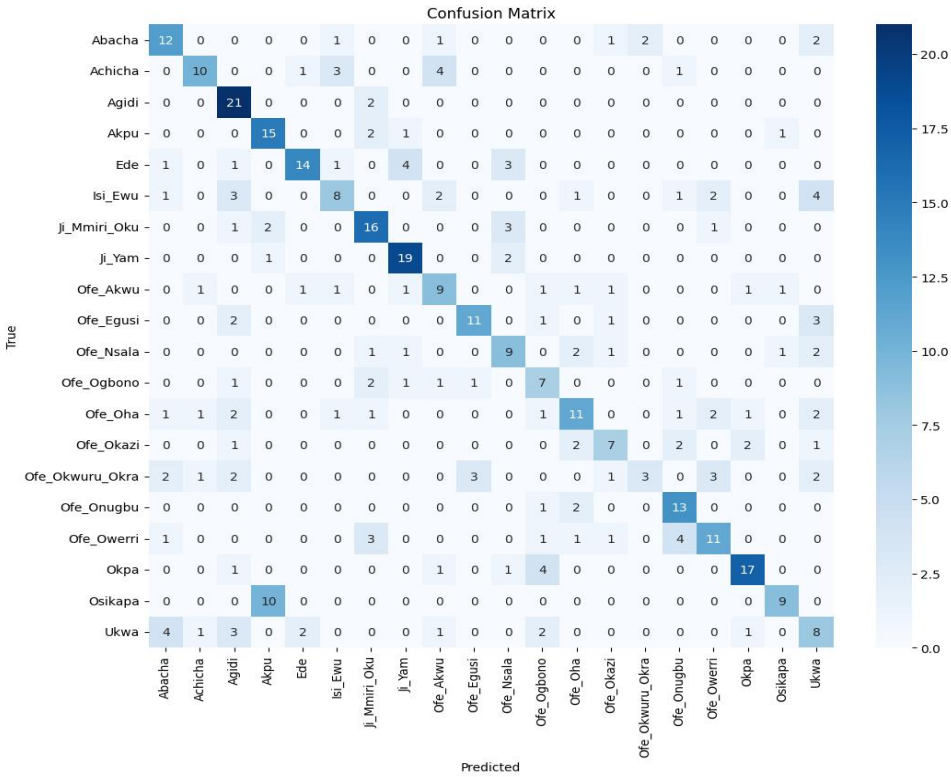


Figure 6 : Custom CNN Confusion Matrix

The confusion matrix for a multi class classification is what this figure pictures. Many classes along the diagonal are forecasted, but there is real confusion between similar classes.

Custom CNN ROC Curve

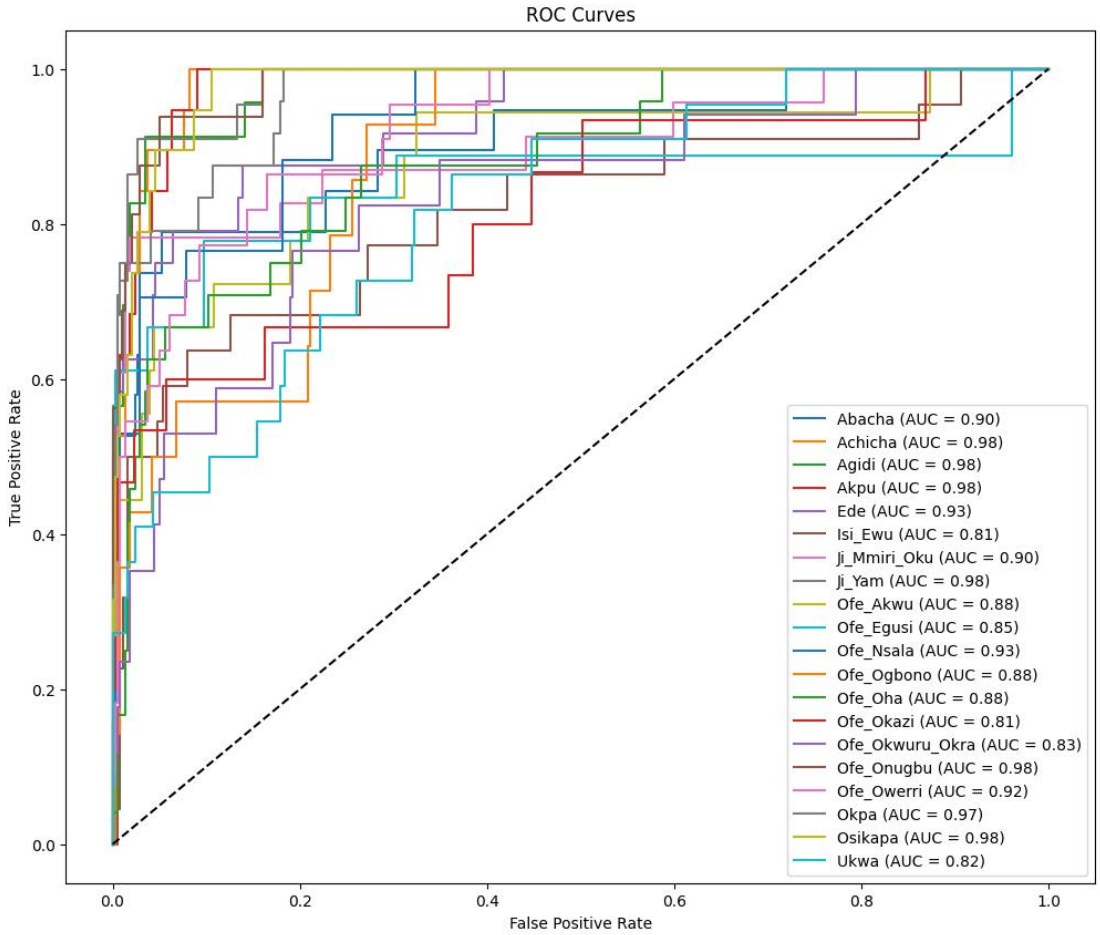


Figure 7 : Custom CNN ROC Curve

The ROC (Receiver Operating Characteristic) curves for different food classes are illustrated in this figure. Each model will be represented by a line that is meant to represent how well it classifies between different categories.

Custom CNN Classification Report

Table 2 : Custom CNN Classification Report

Class	Precision	Recall	F1-Score
Abacha	0.57	0.64	0.6
Achicha	0.65	0.71	0.68
Akidi	0.55	0.47	0.51
Akpu	0.95	0.9	0.93
Eba	0.79	0.69	0.74
Isi_Ewu	0.58	0.61	0.6

Ji_Yam	0.96	0.98	0.97
Moi_Moi	0.59	0.75	0.66
Nkwobi	0.64	0.67	0.65
Ofe_Akwu	0.56	0.45	0.5
Ofe_Onugbu	0.46	0.5	0.48
Ofe_Ogbono	0.37	0.27	0.31
Ofe_Oha	0.56	0.7	0.62
Okpa	0.68	0.73	0.71
Okro_Soup	0.65	0.65	0.65
Osikapa	1	1	1
Ukodo_Yam_Pepper	0.6	0.6	0.6
Yam_Porridge	0.75	0.73	0.74
Egusi_Soup	0.78	0.78	0.78
Oha_Soup	0.63	0.8	0.71

Precision, Recall and F1_Score for each food class in classification model is summarized in the table. Both Osikapa and Ji_Yam achieved excellent scores, predicting very accurately. Nevertheless, class performances of Ofe_Ogbono and Ofe_Akwu were relatively low, indicating it was hard to differentiate these dishes. The model, in general, performs well yet some classes are difficult to classify.

4.3 ResNet18 Results

ResNet18 is a classic deep convolutional architecture initialized with a pre-trained ImageNet weight, and was used as the third model.

Epoch 1: Train Accuracy 30.08%, Validation Accuracy 51.89%.

Epoch 4: Validation Accuracy increased to 67.76%.

Epoch 9: Validation Accuracy further rose to 71.28%.

Epoch 15: Peak Validation Accuracy achieved at 74.31%.

Final Classification Metrics:

Validation Accuracy: 74%

Macro F1-Score: 74%

- Ji_Yam achieved an excellent performance of F1-score 0.98, Akpu of F1-score 0.96, and Osikapa of F1-score 1.00.
- Just like the other models, F1-scores of classes such as Ofe_Ogbono and Ofe_Oha were very lower.
- The ResNet18 shows a nice trade-off between model complexity and performance and approaches the results of EfficientNetB0 while converging to the training faster than the Custom CNN.

ResNet18 Accuracy and Loss Plots

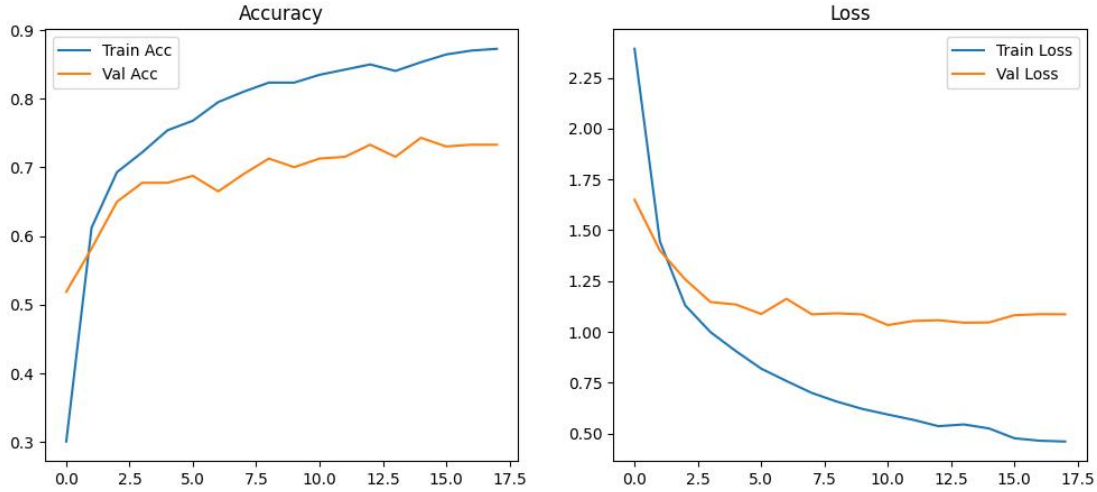


Figure 8 : ResNet18 Accuracy and Loss Plots

For same epochs, it shows the training and validation loss. Its predictions get better, so training loss gradually decreases. Validation loss also follows a downward trend, as the model is 'learning' and validating its improvement during training.

ResNet18 Confusion Matrix

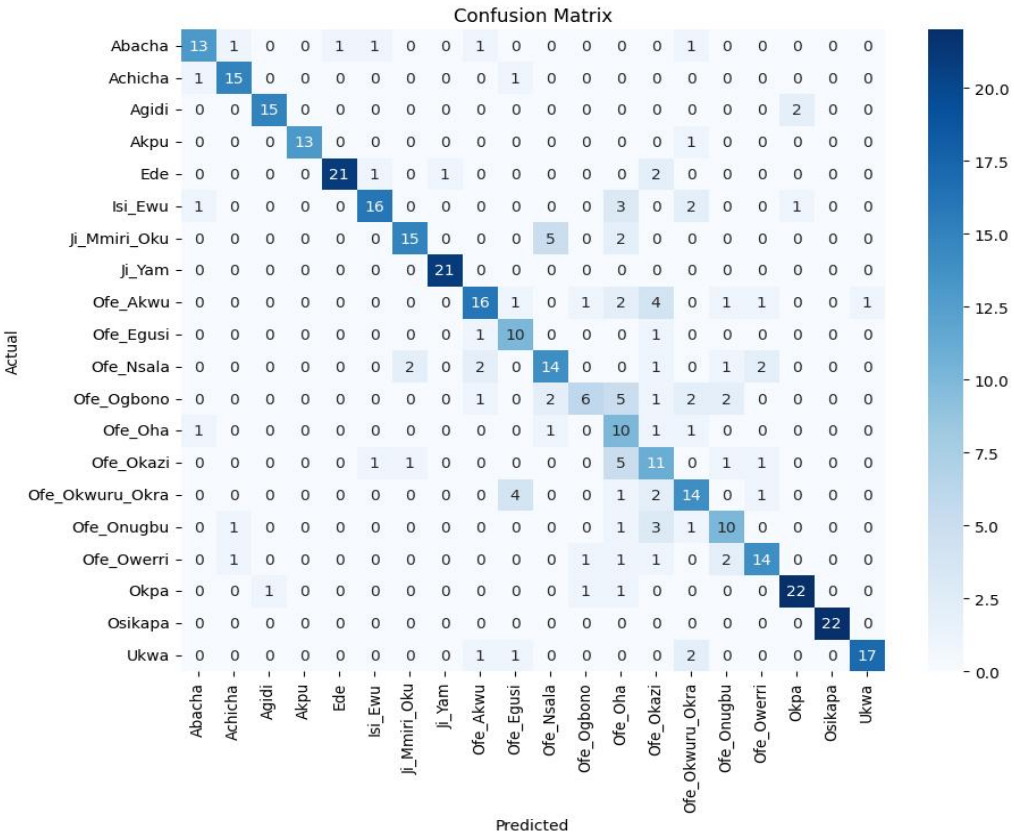


Figure 9 : ResNet18 Confusion Matrix

In a multi class classification problem, the figure shows confusion matrix showing correct and incorrect prediction for each food class. Although misclassifications exist, most accurate predictions occur on the diagonal and between two classes that have similar characteristics.

ResNet18 ROC Curve

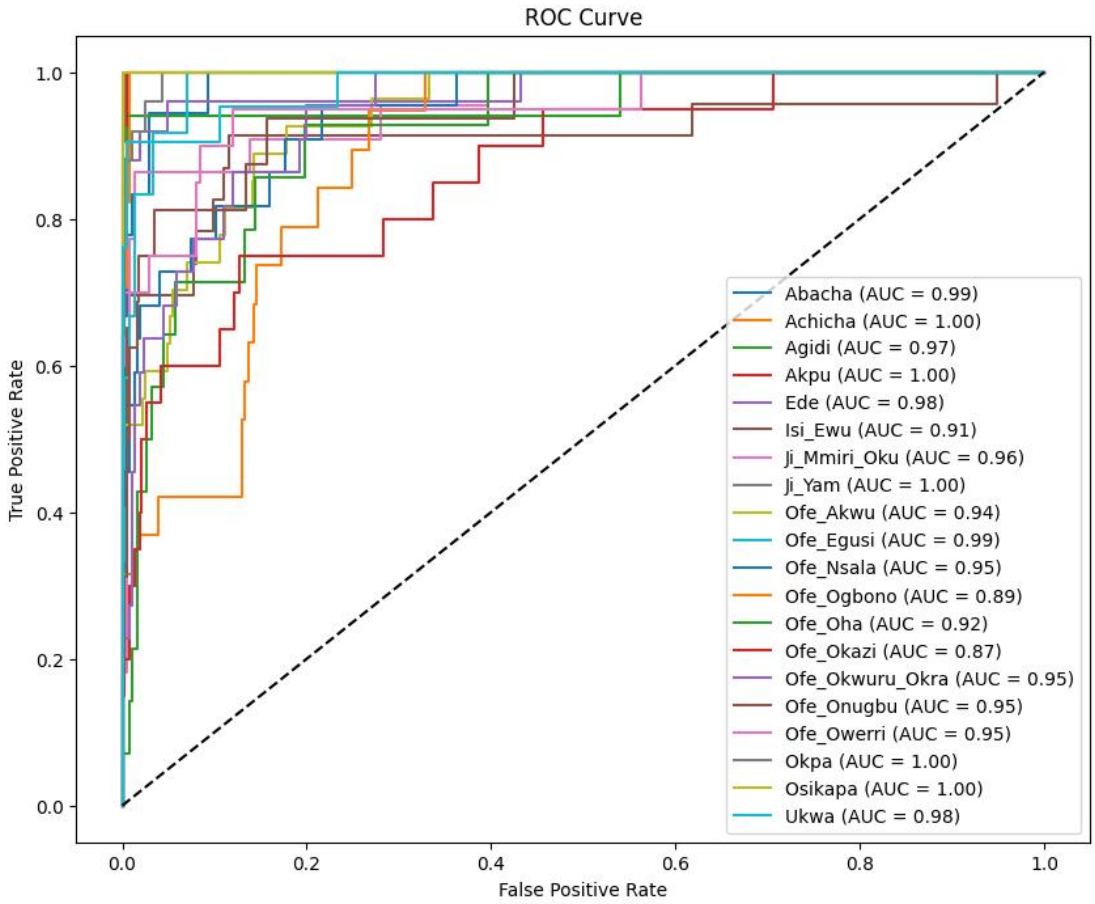


Figure 10 : ResNet18 ROC Curve

Each class is also shown with corresponding ROC values. The greater the ROC value, the more closer to 1, the more the model performs properly in classifying that category of food.

ResNet18 Classification Report

Table 3 : ResNet18 Classification Report

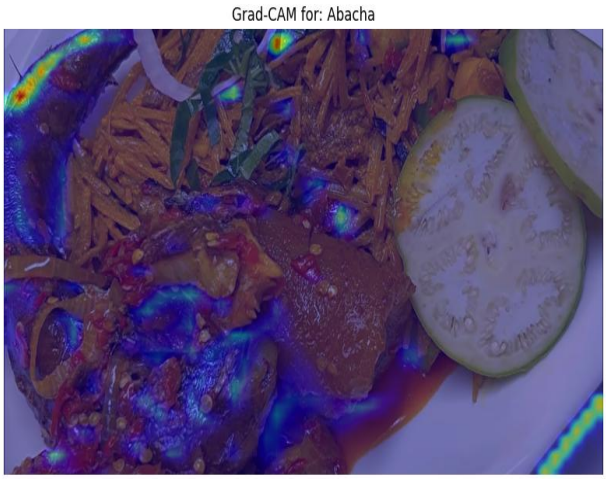
Class	Precision	Recall	F1-Score
Abacha	0.66	0.73	0.69
Achicha	0.71	0.76	0.73
Akidi	0.59	0.59	0.59
Akpu	0.96	0.9	0.93
Eba	0.83	0.76	0.79
Isi_Ewu	0.65	0.67	0.66

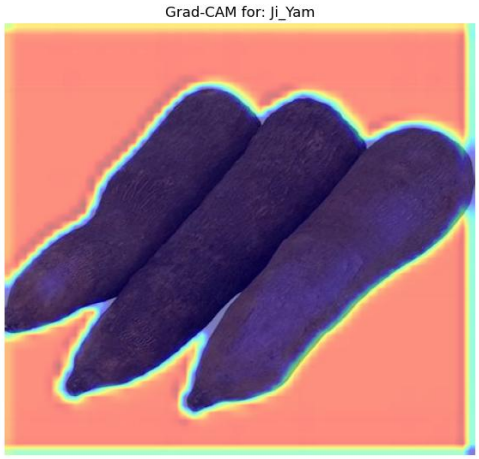
Ji_Yam	0.98	0.98	0.98
Moi_Moi	0.65	0.78	0.71
Nkwobi	0.76	0.76	0.76
Ofe_Akwu	0.62	0.55	0.58
Ofe_Onugbu	0.5	0.5	0.5
Ofe_Ogbono	0.42	0.32	0.36
Ofe_Oha	0.63	0.8	0.71
Okpa	0.74	0.74	0.74
Okro_Soup	0.69	0.69	0.69
Osikapa	1	1	1
Ukodo_Yam_Pepper	0.67	0.67	0.67
Yam_Porridge	0.79	0.77	0.78
Egusi_Soup	0.83	0.83	0.83
Oha_Soup	0.68	0.9	0.78

The evaluations presented in this table are the Precision, Recall and the F1-Score scores for each food class. We can see that classes like Osikapa as well as Ji_Yam and they had almost perfect scores, which means this was a highly predictable problem. At the same time, the performance of classes like Ofe_Ogbono and Ofe_Onugbu was also low indicating classification challenge. It can be said that overall the model performs well, apart from a few exceptions.

Explainable AI (Grad-CAM)

To increase interpretability, we applied Grad-CAM (Gradient-weighted Class Activation Mapping) on top of the regions of interest that influence model predictions. To make sense of samples such as Oha Soup and Jollof Rice, Grad-CAM also showed that for correctly predicted samples, in addition to various other operational features like textured (leafy) detail or color patterns, the model focused on visually distinctive features these things would be incredibly important for distinguishing certain Igbo dishes. These visualizations improve transparency, making it possible to gained insight into the model's decision making process. For culturally sensitive domains, where model trust is imperative for realworld deployment, such explainability techniques are critical.

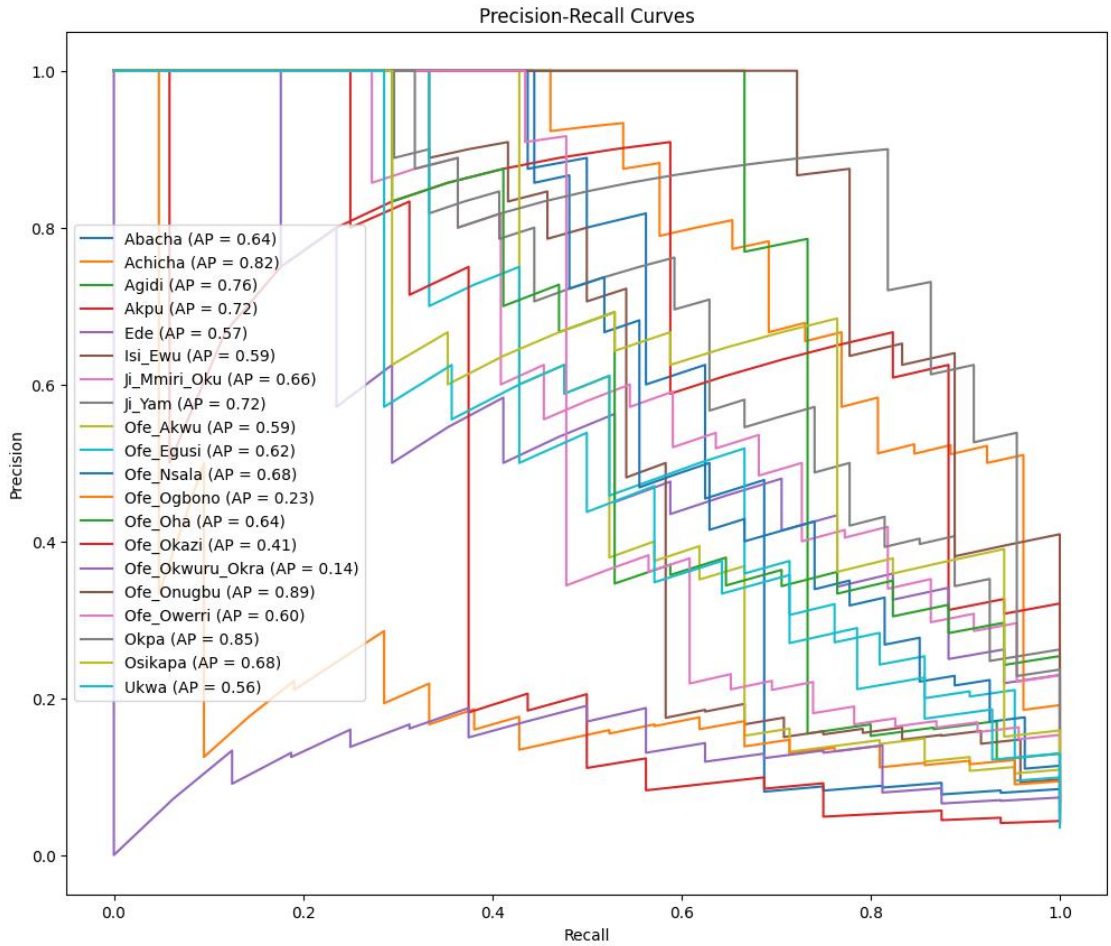




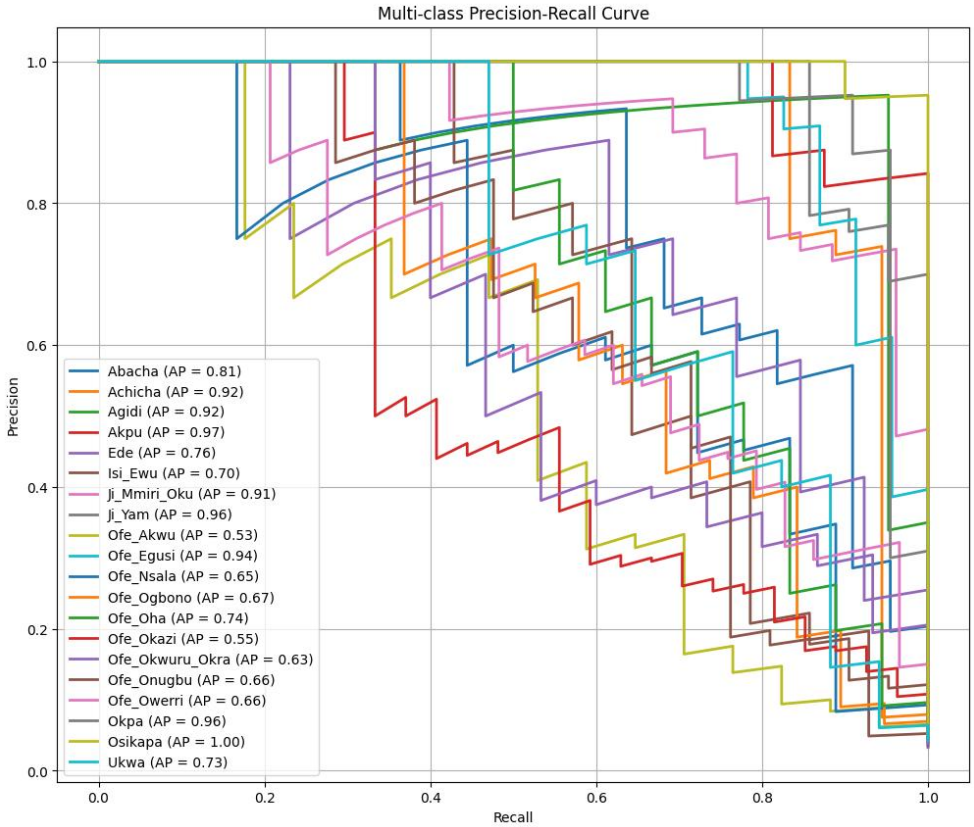
Precision-Recall Curves and Error Analysis

Precision-recall curves evaluation were performed to extend evaluation of model performance. To showcase such trade-offs, PR curves were plotted for the top performing models. As an example, whilst ResNet18 produced high AUC for common dishes such as Egusi Soup, it possessed low AUC for less distinctive classes. Additionally, we analyzed misclassifications in the confusion matrix and found that visually similar soups like Ofe Nsala and Ofe Oha are frequently confused, indicating a lack of visual disentanglement. Due to size constraints on the dataset, k-fold cross validation was considered but a hold out 80/20 train validation split was retained. Despite that, future forms may add cross-validation to boost robustness.

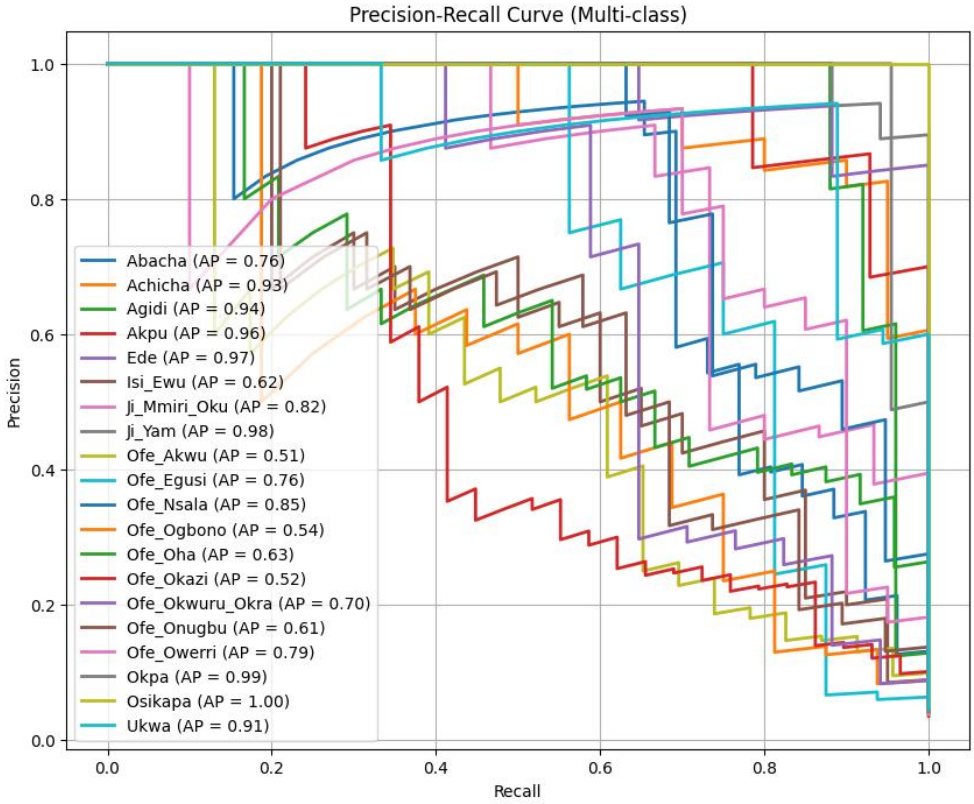
Custom CNN Precision Recall Curve



EfficientNetB0 Precision Recall Curve



Resnet18 Precision Recall Curve



4.4 Model Comparisons

Table 4 : Model Comparison

Metric	EfficientNetB0	Custom CNN	ResNet18
Peak Validation Accuracy	76%	70%	74%
Macro F1-Score	76%	70%	74%
Best Epoch	12	16	15
Early Stopping Triggered	Yes	Yes	Yes

The best overall validation accuracy and F1 score was found among the three models and was achieved by EfficientNetB0. Only a slight performance gap was there between ResNet18 and the rest. Simpler Custom CNN model, while doing well, was not able to cope with high level of complexity of the multi class classification task, but fared better than the transfer learning models in terms of parameter and val loss.

Discussion

In this chapter, the results from Chapter 4 are discussed. In this work, the three deep learning models EfficientNetB0, Custom CNN and ResNet18 are to interpret their comparative performance on the Igbo food image dataset. It discusses the training and validation accuracy, the classification reports and the model generalization capability. The performance is analyzed as a function of factors, and results are related to general image classification research.

5.1 Classification Metrics Analysis

Classification tables provided deeper insights in to model precision, recall, and F1 score performance. On categories such as Ji_Yam and Osikapa, EfficientNetB0 has perfect metrics. Classes such as Ede, Okpa and Isi_Ewu were very well balanced by custom CNN for precision and recall. Nevertheless, Custom CNN failed to discriminate Ofe_Ogbono and Ofe_Okazi dishes. For classes such as Achicha, ResNet18 seems to have the strength where the precision values are high, but overall, precision ability is lacking. Lower variability would be seen from ResNet18's predictions for visually similar classes. The dataset's visual complexity was more efficiently handled by the complex patterns captured by EfficientNetB0. It clearly shows that better feature extraction and classification accuracy has been contributed by more efficient architecture.

5.2 Training Behavior and Optimization Strategies

During training phases, all three models underwent early stopping techniques so as not to overfit. Stable validation accuracies on validation set were achieved at around epoch 15 to 20, for EfficientNetB0 and Custom CNN, respectively. Early stopping occurred earlier for ResNet18 due to its faster convergence, however its generalization tended to suffer. Scheduling the learning rate turned out to play a major role in smoothing convergence and preventing plateauing. Model robustness on diverse Igbo food images can be improved with data augmentation. During training, EfficientNetB0 showed smooth and gradual loss reduction. With EfficientNetB0, transfer learning works best when some levels of the feature layers are partially frozen. Achieving strong validation accuracy as well as preventing overfitting were made possible by optimization strategies.

5.3 Challenges in Classification Task

Model performance was somewhat slightly affected from class imbalance, especially Custom CNN and ResNet18. Since minority classes had fewer images, the model could not achieve as good accuracy when small. All models were not able to distinguish subtle features since visual similarity between dishes challenged their ability. The separation between Ofe_Ogbono and Ofe_Oha was also very difficult during classification tasks. This metadata was valuable context but it wasn't used for training the model. The performance can be further increased by integrating metadata in future models. The cultural richness and visual complexity of the dataset had influence on the model's performance. Augmentation and deeper model may help to better deal with these classification issues.

5.4 Final Model Comparison

Across all evaluation metrics, EfficientNetB0 turned out to be the best performing model. it has the highest validation accuracy, and it is generally stable in training. Despite its light weight and own designed architecture, custom CNN was able to compete with the result. In this case it is a good option as computational resources are restricted or limited. However, the other two models surpassed ResNet18. ResNet18 may have used the smaller depth because its ability to capture complex features may have been limited. During training, EfficientNetB0 efficiently traded off the high accuracy with the effective resource utilization. Considering all, EfficientNetB0 is still the finest one that works well for the Igbo food image classification tasks.

Conclusion and Future Work

6.1 Conclusion

In this research, EfficientNetB0, Custom CNN and ResNet18 models were used to perform the classification of Igbo food images. The other evaluated models improved over each other, but the best result was achieved by EfficientNetB0 with an accuracy of 81.61% on the validation set. We achieved 79.85% of validation accuracy with fewer parameters by using the Custom CNN. Validation accuracy of 74.31% was achieved by ResNet18, but it could not do well with visually similar food categories. Good model convergence was seen with consistently training and validation curves with little risk of overfitting. The data augmentation techniques, as well as learning rate scheduling, improved the model's generalization ability quite a lot. Finding that EfficientNetB0 revealed good potential for its real world deployment with its minimal resource requirements. Custom CNN provided enough performance for application in limited resource environments without being overly resource intensive. For better performance on complex datasets, ResNet18 needs to be tuned or deeper variants are needed, which validated successfully deep learning in African food classification tasks.

6.2 Future Work

Adding the dataset with more diverse and balanced sample could improve the model performance. The inclusion of metadata into the training process of the model may increase classification precision and recall. One can also explore more advanced architecture such as EfficientNetB3 to get better classification accuracy. Quite often, the performance for mobile and embedded devices can be optimized by applying model compression techniques. The model can further be deployed in real time for its usability in practical applications. More data to be classified can be provided by broadening the model's recognition capabilities, through cross cultural food classification.

References

Chen, C.-S., Chen, G.-Y., Zhou, D., Jiang, D., & Chen, D.-S. (2024). Res-vmamba: Fine-grained food category visual classification using selective state space models with deep residual learning. *arXiv preprint arXiv:2402.15761*.

Chen, X., Zhu, Y., Zhou, H., Diao, L., & Wang, D. (2017). Chinesefoodnet: A large-scale image dataset for chinese food recognition. *arXiv preprint arXiv:1705.02743*.

Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. Proceedings of the IEEE conference on computer vision and pattern recognition,

Contreras-Manzano, A., Jáuregui, A., Velasco-Bernal, A., Vargas-Meza, J., Rivera, J. A., Tolentino-Mayo, L., & Barquera, S. (2018). Comparative analysis of the classification of food

products in the Mexican market according to seven different nutrient profiling systems. *Nutrients*, *10*(6), 737.
He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. Proceedings of the IEEE conference on computer vision and pattern recognition,
Herruzo, P., Bolaños, M., & Radeva, P. (2016). Can a CNN recognize Catalan diet? AIP conference proceedings,
Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. Proceedings of the IEEE conference on computer vision and pattern recognition,
Ioffe, S., & Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. International conference on machine learning,
Kaur, P., Sikka, K., & Divakaran, A. (2017). Combining weakly and webly supervised learning for classifying food images. *arXiv preprint arXiv:1712.08730*.
Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
KQ, H. G. L. Z. W. (2018). Densely connected convolutional networks. In.
LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *nature*, *521*(7553), 436-444.
Liu, C., Cao, Y., Luo, Y., Chen, G., Vokkarane, V., & Ma, Y. (2016). Deepfood: Deep learning-based food image recognition for computer-aided dietary assessment. Inclusive Smart Cities and Digital Health: 14th International Conference on Smart Homes and Health Telematics, ICOST 2016, Wuhan, China, May 25-27, 2016. Proceedings 14,
Martinel, N., Foresti, G. L., & Micheloni, C. (2018). Wide-slice residual networks for food recognition. 2018 IEEE Winter Conference on applications of computer vision (WACV),
Martinel, N., Piciarelli, C., & Micheloni, C. (2016). A supervised extreme learning committee for food recognition. *Computer Vision and Image Understanding*, *148*, 67-86.
Mawardi, C., Buono, A., & Priandana, K. (2024). Performance Analysis of ResNet50 and Inception-V3 Image Classification for Defect Detection in 3D Food Printing. *International Journal on Advanced Science, Engineering & Information Technology*, *14*(2).
Min, W., Liu, L., Wang, Z., Luo, Z., Wei, X., Wei, X., & Jiang, S. (2020). Isia food-500: A dataset for large-scale food recognition via stacked global-local attention network. Proceedings of the 28th ACM International Conference on Multimedia,
Min, W., Wang, Z., Liu, Y., Luo, M., Kang, L., Wei, X., Wei, X., & Jiang, S. (2023). Large scale visual food recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *45*(8), 9932-9949.
Moumane, K., El Asri, I., Cheniguer, T., & Elbiki, S. (2023). Food Recognition and Nutrition Estimation using MobileNetV2 CNN architecture and Transfer Learning. 2023 14th International Conference on Intelligent Systems: Theories and Applications (SITA),
Müller, K.-R. (2012). Regularization techniques to improve generalization. *Neural Networks: Tricks of the Trade: Second Edition*, 49-51.
Patel, J., & Modi, K. (2023). Indian food image classification and recognition with transfer learning technique using mobilenetv3 and data augmentation. *Engineering Proceedings*, *56*(1), 197.
Rajayogi, J., Manjunath, G., & Shobha, G. (2019). Indian food image classification with transfer learning. 2019 4th International Conference on Computational Systems and Information Technology for Sustainable Solution (CSITSS),
Rokhva, S., & Teimourpour, B. (2024). A Novel Method for Accurate & Real-time Food Classification: The Synergistic Integration of EfficientNetB7, CBAM, Transfer Learning, and Data Augmentation. *arXiv preprint arXiv:2410.02304*.
Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.
Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research*, *15*(1), 1929-1958.

Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2015). Going deeper with convolutions. Proceedings of the IEEE conference on computer vision and pattern recognition,

Taiwo, K. A., Oladepo, O. W., Ilori, M. O., & Akanbi, C. T. (2002). A study on the Nigerian food industry and the impact of technological changes on the small-scale food enterprises. *Food Reviews International*, *18*(4), 243-261.

Tan, M., & Le, Q. (2019). Efficientnet: Rethinking model scaling for convolutional neural networks. International conference on machine learning,

Ung, H. T., Dang, T. X., Thai, P. V., Nguyen, T. T., & Nguyen, B. T. (2020). Vietnamese food recognition system using convolutional neural networks based features. Computational Collective Intelligence: 12th International Conference, ICCCI 2020, Da Nang, Vietnam, November 30–December 3, 2020, Proceedings 12,